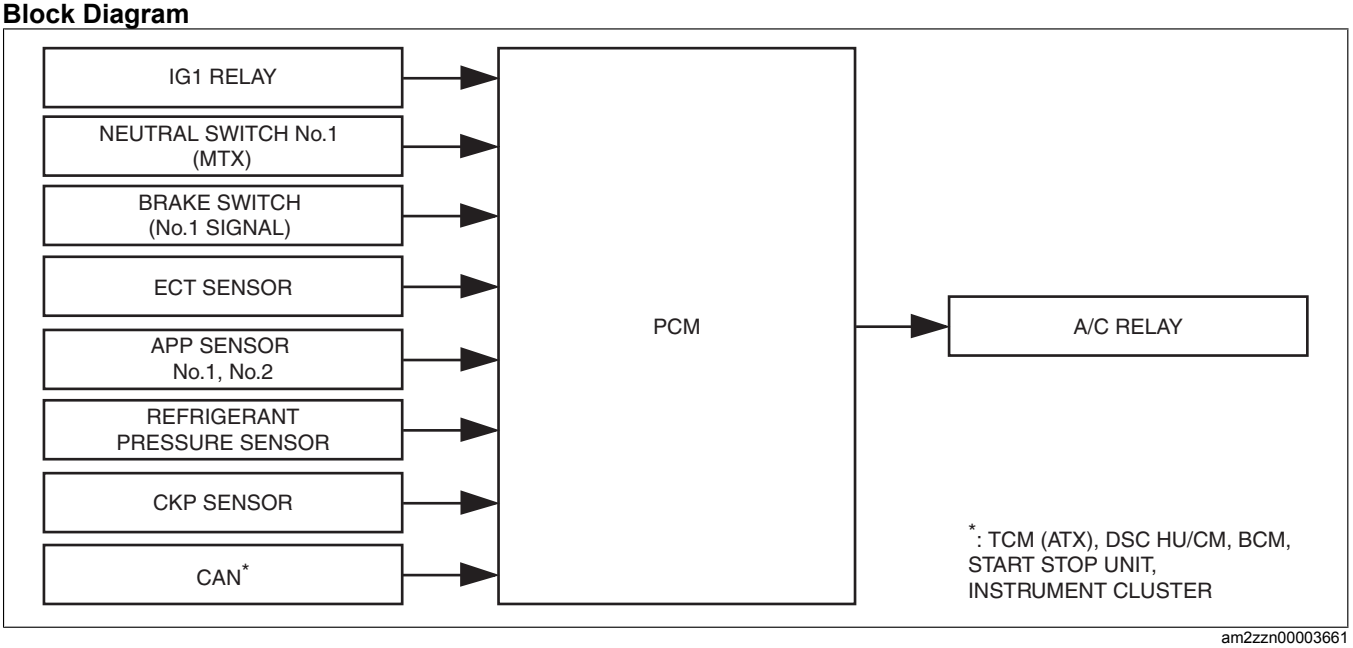
## A/C CUT-OFF CONTROL [SKYACTIV-D 1.5]

id0140z5206200

### Outline

- Controls the A/C operation by switching the A/C relay on/off at the optimal timing according to engine operation conditions. By controlling the A/C operation, acceleration performance and engine reliability have been improved.

### Block Diagram




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### Operation

- The PCM stops energization to the A/C relay when any of the following conditions is met:

### A/C cut-off control operation conditions

—: Not applicable

| Operation condition | A/C relay non-energization time | Purpose |
|---|---|---|
| At engine start | **Approx. 4 s** | To improve startability |
| At engine restart by i-stop control | **Approx. 3 s** | To improve startability |
| Accelerator pedal opening angle **60 % or more** (at acceleration) | **Approx. 5 s** | To improve acceleration performance |
| At i-stop control switching | — | To improve i-stop function reliability |
| Refrigerant pressure **3.14 MPa {32.0 kgf/cm$^2$, 455 psi} or more** | Until refrigerant pressure decreases to **2.55 MPa {26.0 kgf/cm$^2$, 370 psi} or less** | To assure A/C compressor reliability |
| Refrigerant pressure of **0.196 MPa {2.00 kgf/cm$^2$, 28.4 psi} or less** continues for **5 s or more** | Until refrigerant pressure of **0.226 MPa {2.30 kgf/cm$^2$, 32.8 psi} or more** continues to **5 s or more** | To assure A/C compressor reliability |
| Refrigerant pressure sensor malfunction | — | To assure A/C compressor reliability |
| Panic braking determined | — | To assure load performance |
| ECT sensor malfunction | — | Engine protection |
| Engine coolant temperature **113 °C {235 °F} or more** | ON and OFF **every 10 s** until engine coolant temperature lowers to **less than 110 °C {230 °F}** | Engine protection |
| Engine coolant temperature **116 °C {244 °F} or more** | Until engine coolant temperature lowers to less than approx. 113 °C {235 °F} | Engine protection |
| During DPF regeneration control (compulsory DPF regeneration control) | Until DPF regeneration control is completed | To stabilize DPF regeneration control |
| During fuel injection amount learning (manual) | Until fuel injection amount learning is completed | To stabilize fuel injection amount learning |